LABORATORY RECORD | UNIT 5

Python Programming

Lab Experiments - Part 5

Object-Oriented Programming | Simulation | Random Module

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EXPERIMENTS COVERED

14 Bank Account System using OOP

15 Library Management System

16 Dice Simulator using the random module

Every program in this document was executed and the output shown is the real output.

Contents and Approach

This document solves the three laboratory exercises given for Unit 5. For each exercise the same structure is followed so that every point asked in the question is answered clearly and in order. The original question is placed in a highlighted box at the start of each exercise.

No.	Exercise	Main concepts
14	Bank Account System using OOP	Classes, objects, encapsulation, inheritance, custom exceptions, validation
15	Library Management System	Inheritance, polymorphism, availability tracking, issue / return, fines
16	Dice Simulator (random module)	randint, seed, Counter, frequency analysis, probability, board games

Structure followed in every exercise

- **Question** - the exact task statement, highlighted.
- **Aim and concepts used** - what is being built and which Python ideas are applied.
- **Design** - classes, attributes, methods and the algorithm in plain steps.
- **Program** - the complete, runnable Python code with line numbers and colour highlighting.
- **Output** - the real terminal output captured by running the program.
- **Analysis** - test-case table, explanation of results and the discussion the question asks for.
- **Requirement checklist and conclusion** - every sentence of the question mapped to where it is solved.

How to run the programs

Each program is a single file that needs only the Python standard library (Python 3.8 or newer). Save it and run it from a terminal:

TERMINAL

```
python ex14_bank.py
python ex15_library.py
python ex16_dice.py           # demonstration run
python ex16_dice.py --play    # optional interactive dice roller
```

The dice program uses `random.seed(2026)` so the demonstration output is reproducible. Remove that line to get fresh random numbers on every run. The library program uses fixed dates so the due dates and fines are also reproducible.

14 Bank Account System using OOP

Lab Exercise 14 | Classes, objects, encapsulation, validation

QUESTION

Bank Account System Using OOP: Develop a Python program using classes and objects to manage bank accounts. Implement deposit, withdrawal, balance enquiry, and transaction validation.

- Design appropriate classes and methods, apply encapsulation, analyze transaction conditions, and test invalid transactions and insufficient balances.
- Simulate routine banking transactions and explain how object-oriented programming supports account management and data protection.

1. Aim

To build an object-oriented banking system in Python in which every account is an object that protects its own data, supports deposit, withdrawal and balance enquiry, validates every transaction, and rejects invalid requests and requests that exceed the available balance.

2. Concepts used

Concept	How it is used in the program
Class and object	**BankAccount**, **SavingsAccount** and **Bank** are classes; every customer account is an object.
Encapsulation	Balance, PIN, status and history are **private** (`__balance`). They can only change through methods that check the rules.
Inheritance	**SavingsAccount** inherits everything from BankAccount and adds monthly interest.
Composition	**Bank** holds many account objects and handles transfers between them.
Custom exceptions	InvalidAmountError, InsufficientFundsError, AccountInactiveError, all derived from BankError.
Class attributes	MIN_BALANCE (500) and INTEREST_RATE (4%) are shared by all objects of the class.
Static method	**_validate_amount()** needs no object, so it is a staticmethod.

3. Class design

Class	Important attributes	Important methods
BankAccount	acc_no, holder (public); __balance, __pin, __active, __history (private)	deposit(), withdraw(), get_balance(), mini_statement(), close(), _validate_amount(), _check_pin()
SavingsAccount (child)	INTEREST_RATE = 4.0 (adds to the parent)	add_interest()
Bank	name, __accounts (dictionary)	open_account(), find(), transfer(), total_deposits()
BankError (exceptions)	-	InvalidAmountError, InsufficientFundsError, AccountInactiveError

4. Algorithm

Deposit

1. Check that the account is active.
2. Validate the amount: it must be a number greater than zero. Otherwise raise InvalidAmountError.
3. Add the amount to the private balance.

4. Record the transaction in the history and return the new balance.

Withdrawal

1. Check that the account is active and that the PIN is correct.
2. Validate the amount (positive number).
3. Compute balance - amount. If the result is below MIN_BALANCE (500), raise InsufficientFundsError.
4. Otherwise subtract the amount, record the transaction and return the new balance.

Transfer

- Find both accounts, call withdraw() on the sender (so all rules above are re-used automatically), then credit the receiver.

5. Program

ex14_bank.py

```python
"""
Lab Exercise 14 : Bank Account System using OOP
Concepts        : Classes, objects, encapsulation, inheritance,
                  custom exceptions, transaction validation
"""

from datetime import datetime


# ------------------------------------------------------------
# Custom exceptions - make error handling meaningful and readable
# ------------------------------------------------------------
class BankError(Exception):
    """Base class for all banking related errors."""


class InvalidAmountError(BankError):
    """Raised when the amount is not a positive number."""


class InsufficientFundsError(BankError):
    """Raised when a withdrawal exceeds the available balance."""


class AccountInactiveError(BankError):
    """Raised when a transaction is tried on a closed account."""


# ------------------------------------------------------------
# Class 1 : BankAccount  (encapsulation - balance is PRIVATE)
# ------------------------------------------------------------
class BankAccount:
    MIN_BALANCE = 500          # class attribute: minimum balance to be maintained
    _next_txn_id = 1           # class attribute: auto-incrementing transaction id

    def __init__(self, acc_no, holder, opening_balance=0.0, pin="0000"):
        if opening_balance < 0:
            raise InvalidAmountError("Opening balance cannot be negative.")
        self.acc_no = acc_no                    # public
        self.holder = holder                    # public
        self.__balance = float(opening_balance) # private (name-mangled)
        self.__pin = str(pin)                   # private
        self.__active = True                    # private
        self.__history = []                     # private transaction log
        self._log("OPEN", opening_balance, "Account opened")

    # ----- private helpers (not meant to be used outside the class) -----
    def _log(self, kind, amount, note=""):
        entry = {
            "id": BankAccount._next_txn_id,
            "time": datetime.now().strftime("%H:%M:%S"),
            "type": kind,
            "amount": float(amount),
            "balance": self.__balance,
            "note": note,
        }
        BankAccount._next_txn_id += 1
        self.__history.append(entry)

```

```
    @staticmethod
    def _validate_amount(amount):
        """Transaction validation: amount must be a positive number."""
        if isinstance(amount, bool) or not isinstance(amount, (int, float)):
            raise InvalidAmountError(f"Amount must be a number, got {type(amount).__name__}.")
        if amount <= 0:
            raise InvalidAmountError("Amount must be greater than zero.")

    def _check_active(self):
        if not self.__active:
            raise AccountInactiveError("Account is closed. No transactions allowed.")

    def _check_pin(self, pin):
        if str(pin) != self.__pin:
            raise BankError("Incorrect PIN. Transaction denied.")

    # ----- public interface -----
    def deposit(self, amount):
        self._check_active()
        self._validate_amount(amount)
        self.__balance += amount
        self._log("DEPOSIT", amount)
        return self.__balance

    def withdraw(self, amount, pin):
        self._check_active()
        self._check_pin(pin)
        self._validate_amount(amount)
        if self.__balance - amount < self.MIN_BALANCE:
            raise InsufficientFundsError(
                f"Insufficient balance. Available: {self.__balance:.2f}, "
                f"requested: {amount:.2f}, minimum to maintain: {self.MIN_BALANCE}."
            )
        self.__balance -= amount
        self._log("WITHDRAW", amount)
        return self.__balance

    def get_balance(self):
        """Balance enquiry - read-only access to the private balance."""
        self._check_active()
        return self.__balance

    def mini_statement(self, last_n=5):
        return self.__history[-last_n:]

    def close(self):
        self.__active = False

    def _apply_internal_credit(self, amount, note):
        """Used only by the Bank class for transfers (protected by convention)."""
        self.__balance += amount
        self._log("TRANSFER-IN", amount, note)

    def __str__(self):
        return f"[{self.acc_no}] {self.holder:<10} | Balance: Rs. {self.__balance:>10,.2f}"


# ------------------------------------------------------------------
# Class 2 : SavingsAccount (inheritance + method extension)
# ------------------------------------------------------------------
class SavingsAccount(BankAccount):
    INTEREST_RATE = 4.0   # percent per annum

    def add_interest(self):
        interest = self.get_balance() * self.INTEREST_RATE / 100 / 12   # one month
        self.deposit(round(interest, 2))
        return round(interest, 2)


# ------------------------------------------------------------------
# Class 3 : Bank (manages many accounts - composition)
# ------------------------------------------------------------------
class Bank:
    def __init__(self, name):
        self.name = name
        self.__accounts = {}

    def open_account(self, account):
        if account.acc_no in self.__accounts:
            raise BankError("Account number already exists.")
        self.__accounts[account.acc_no] = account

    def find(self, acc_no):
```

```
        if acc_no not in self.__accounts:
            raise BankError(f"Account {acc_no} not found.")
        return self.__accounts[acc_no]

    def transfer(self, src_no, dst_no, amount, pin):
        src, dst = self.find(src_no), self.find(dst_no)
        src.withdraw(amount, pin)        # re-uses all validation rules
        # withdraw() already logged a WITHDRAW entry; credit the receiver
        dst._apply_internal_credit(amount, f"From {src_no}")

    def total_deposits(self):
        return sum(a.get_balance() for a in self.__accounts.values())


# ------------------------------------------------------------------
# Test harness - tries a transaction and reports PASS / BLOCKED
# ------------------------------------------------------------------
def attempt(title, action):
    try:
        result = action()
        print(f"  [OK]      {title:<42} -> {result}")
    except BankError as err:
        print(f"  [BLOCKED] {title:<42} -> {type(err).__name__}: {err}")
    except Exception as err:                      # unexpected errors
        print(f"  [ERROR]   {title:<42} -> {type(err).__name__}: {err}")


def main():
    bank = Bank("PyBank")
    a1 = SavingsAccount("SB1001", "Sriram", 10000, pin="1234")
    a2 = BankAccount("CA2002", "Rahul", 2500, pin="4321")
    bank.open_account(a1)
    bank.open_account(a2)

    print("=" * 78)
    print(" 1. ROUTINE TRANSACTIONS")
    print("=" * 78)
    print(a1)
    print(a2)
    attempt("Deposit 5000 into SB1001", lambda: a1.deposit(5000))
    attempt("Withdraw 2000 from SB1001 (PIN ok)", lambda: a1.withdraw(2000, "1234"))
    attempt("Balance enquiry SB1001", lambda: a1.get_balance())
    attempt("Add monthly interest to SB1001", lambda: a1.add_interest())
    attempt("Transfer 1000 SB1001 -> CA2002", lambda: bank.transfer("SB1001", "CA2002", 1000, "1234"))
    attempt("Balance enquiry CA2002", lambda: a2.get_balance())

    print()
    print("=" * 78)
    print(" 2. INVALID TRANSACTION TESTS")
    print("=" * 78)
    attempt("Deposit negative amount (-300)", lambda: a1.deposit(-300))
    attempt("Deposit zero", lambda: a1.deposit(0))
    attempt("Deposit text 'abc'", lambda: a1.deposit("abc"))
    attempt("Withdraw with wrong PIN", lambda: a1.withdraw(100, "0000"))
    attempt("Withdraw negative amount", lambda: a1.withdraw(-50, "1234"))

    print()
    print("=" * 78)
    print(" 3. INSUFFICIENT BALANCE TESTS")
    print("=" * 78)
    attempt("Withdraw 50000 from SB1001", lambda: a1.withdraw(50000, "1234"))
    attempt("Withdraw 3200 from CA2002 (min bal rule)", lambda: a2.withdraw(3200, "4321"))
    attempt("Transfer 99999 SB1001 -> CA2002", lambda: bank.transfer("SB1001", "CA2002", 99999, "1234"))
    attempt("Transfer to unknown account ZZ9999", lambda: bank.transfer("SB1001", "ZZ9999", 10, "1234"))

    print()
    print("=" * 78)
    print(" 4. DATA PROTECTION (ENCAPSULATION) DEMO")
    print("=" * 78)
    attempt("Direct read  a1.__balance", lambda: a1.__balance)
    attempt("Name-mangled access (not recommended)", lambda: a1._BankAccount__balance)
    print("  Safe, supported way ->  a1.get_balance() =", a1.get_balance())

    print()
    print("=" * 78)
    print(" 5. CLOSED ACCOUNT TEST")
    print("=" * 78)
    print(f"  Total deposits held by {bank.name}: Rs. {bank.total_deposits():,.2f}")
    a2.close()
    attempt("Deposit into closed account CA2002", lambda: a2.deposit(100))

    print()
    print("=" * 78)
```

```
    print(" 6. MINI STATEMENT OF SB1001")
    print("=" * 78)
    print(f"  {'ID':<4}{'TYPE':<14}{'AMOUNT':>12}{'BALANCE':>14}  NOTE")
    for t in a1.mini_statement(10):
        print(f"  {t['id']:<4}{t['type']:<14}{t['amount']:>12,.2f}{t['balance']:>14,.2f}  {t['note']}")



if __name__ == "__main__":
    main()
```

6. Output

TERMINAL OUTPUT - python ex14_bank.py

```
==============================================================================
 1. ROUTINE TRANSACTIONS
==============================================================================
[SB1001] Sriram     | Balance: Rs.  10,000.00
[CA2002] Rahul      | Balance: Rs.   2,500.00
  [OK]      Deposit 5000 into SB1001                  -> 15000.0
  [OK]      Withdraw 2000 from SB1001 (PIN ok)        -> 13000.0
  [OK]      Balance enquiry SB1001                    -> 13000.0
  [OK]      Add monthly interest to SB1001            -> 43.33
  [OK]      Transfer 1000 SB1001 -> CA2002            -> None
  [OK]      Balance enquiry CA2002                    -> 3500.0

==============================================================================
 2. INVALID TRANSACTION TESTS
==============================================================================
  [BLOCKED] Deposit negative amount (-300)            -> InvalidAmountError: Amount must be greater than zero.
  [BLOCKED] Deposit zero                              -> InvalidAmountError: Amount must be greater than zero.
  [BLOCKED] Deposit text 'abc'                        -> InvalidAmountError: Amount must be a number, got str.
  [BLOCKED] Withdraw with wrong PIN                   -> BankError: Incorrect PIN. Transaction denied.
  [BLOCKED] Withdraw negative amount                  -> InvalidAmountError: Amount must be greater than zero.

==============================================================================
 3. INSUFFICIENT BALANCE TESTS
==============================================================================
  [BLOCKED] Withdraw 50000 from SB1001                -> InsufficientFundsError: Insufficient balance. Available:
            12043.33, requested: 50000.00, minimum to maintain: 500.
  [BLOCKED] Withdraw 3200 from CA2002 (min bal rule)  -> InsufficientFundsError: Insufficient balance. Available:
            3500.00, requested: 3200.00, minimum to maintain: 500.
  [BLOCKED] Transfer 99999 SB1001 -> CA2002           -> InsufficientFundsError: Insufficient balance. Available:
            12043.33, requested: 99999.00, minimum to maintain: 500.
  [BLOCKED] Transfer to unknown account ZZ9999        -> BankError: Account ZZ9999 not found.

==============================================================================
 4. DATA PROTECTION (ENCAPSULATION) DEMO
==============================================================================
  [ERROR]   Direct read  a1.__balance                 -> AttributeError: 'SavingsAccount' object has no attribute
            '__balance'
  [OK]      Name-mangled access (not recommended)     -> 12043.33
  Safe, supported way ->  a1.get_balance() = 12043.33

==============================================================================
 5. CLOSED ACCOUNT TEST
==============================================================================
  Total deposits held by PyBank: Rs. 15,543.33
  [BLOCKED] Deposit into closed account CA2002        -> AccountInactiveError: Account is closed. No transactions
            allowed.

==============================================================================
 6. MINI STATEMENT OF SB1001
==============================================================================
  ID  TYPE              AMOUNT       BALANCE  NOTE
  1   OPEN           10,000.00     10,000.00  Account opened
  3   DEPOSIT         5,000.00     15,000.00
  4   WITHDRAW        2,000.00     13,000.00
  5   DEPOSIT            43.33     13,043.33
  6   WITHDRAW        1,000.00     12,043.33
```

7. Analysis of transaction conditions and test results

Opening balances were Rs. 10,000 (SB1001, PIN 1234) and Rs. 2,500 (CA2002, PIN 4321). The minimum balance rule is Rs. 500.

#	Test performed	Condition checked	Result
1	Deposit 5000 into SB1001	Valid positive amount	Accepted. Balance 15,000.00
2	Withdraw 2000 with correct PIN	PIN ok, balance stays above 500	Accepted. Balance 13,000.00
3	Add monthly interest	13,000 x 4% / 12	Accepted. Interest 43.33 credited
4	Transfer 1000 SB1001 to CA2002	Withdraw rules, then credit	Accepted. SB1001 = 12,043.33, CA2002 = 3,500.00
5	Deposit -300	Amount must be > 0	BLOCKED: InvalidAmountError
6	Deposit 0	Amount must be > 0	BLOCKED: InvalidAmountError
7	Deposit 'abc'	Amount must be a number	BLOCKED: InvalidAmountError (got str)
8	Withdraw with wrong PIN	PIN must match	BLOCKED: BankError
9	Withdraw -50	Amount must be > 0	BLOCKED: InvalidAmountError
10	Withdraw 50000 from SB1001	Balance 12,043.33 is too low	BLOCKED: InsufficientFundsError
11	Withdraw 3200 from CA2002	3,500 - 3,200 = 300, below minimum 500	BLOCKED: InsufficientFundsError
12	Transfer 99999	Sender balance too low	BLOCKED: InsufficientFundsError
13	Transfer to unknown account	Account must exist	BLOCKED: BankError
14	Read a1.__balance directly	Private attribute	AttributeError - data protected
15	Deposit into closed account	Account must be active	BLOCKED: AccountInactiveError

Observation: every invalid request was stopped by the object itself before any data changed. After all the blocked attempts the balance of SB1001 was still exactly 12,043.33, which proves that failed transactions leave no side effects.

8. How OOP supports account management and data protection

- **Encapsulation protects data.** The balance is stored in a private attribute. Outside code cannot write `acct.balance = 1000000`; it must call deposit(), which validates the amount first. Test 14 shows that even reading `__balance` directly fails.
- **Abstraction keeps usage simple.** A user of the class only needs deposit(), withdraw() and get_balance(). The PIN check, minimum balance rule and logging are hidden inside the methods.
- **One place for every rule.** Because all changes pass through the same methods, a rule such as the Rs. 500 minimum balance is written once and enforced everywhere, including transfers.
- **Inheritance supports reuse.** SavingsAccount gets all the safe behaviour of BankAccount and only adds interest. A CurrentAccount or LoanAccount could be added the same way.
- **Objects model real accounts.** Each account has its own balance and history, so thousands of accounts can be managed independently inside the Bank object.
- **Exceptions give clear errors.** Different error classes let the calling code react differently to a bad amount, a low balance or a closed account.
- **Limitation to remember:** Python's double underscore only renames the attribute (`_BankAccount__balance`, test shown in the output). It discourages accidental access but is not real security. A real bank also uses databases, encryption and authentication.

9. Requirement checklist

Requirement in the question	Where it is solved
Classes and objects to manage accounts	BankAccount, SavingsAccount, Bank (Sections 3, 5)
Deposit, withdrawal, balance enquiry	deposit(), withdraw(), get_balance()
Transaction validation	_validate_amount(), _check_pin(), _check_active(), minimum balance rule
Apply encapsulation	Private __balance, __pin, __active, __history
Analyse transaction conditions	Section 7 test table
Test invalid transactions and insufficient balances	Output sections 2 and 3
Simulate routine transactions	Output section 1 and mini statement (section 6)
Explain OOP for account management and data protection	Section 8

10. Conclusion

The program successfully manages accounts using classes and objects. Deposits, withdrawals, enquiries and transfers work correctly for valid input, while negative amounts, wrong PINs, non-numeric input, low balances, unknown accounts and closed accounts are all rejected safely. Encapsulation and validation inside the class keep the account data consistent and protected.

15 Library Management System

Lab Exercise 15 | Inheritance, availability tracking, issue and return

QUESTION

Library Management System: Develop a Python program using classes and objects to manage books, issue and return books, and maintain availability status.

- Identify suitable classes and attributes, implement methods, apply inheritance where appropriate, and test book availability and issue/return conditions.
- Simulate library operations and demonstrate how the system supports book tracking and efficient resource management.

1. Aim

To build a library system where books and magazines are objects, members borrow and return them, the number of available copies is updated automatically, and rules such as borrowing limits, duplicate issue and late fines are enforced.

2. Concepts used

Concept	How it is used in the program
Classes and objects	LibraryItem, Book, Magazine, Member and Library; each book copy set is an object.
Inheritance	**Book** and **Magazine** inherit from **LibraryItem**, re-using id, title, copies and status logic.
Method / attribute overriding	Magazine changes LOAN_DAYS to 7 and FINE_PER_DAY to 1; kind() returns a different label.
Polymorphism	The Library calls item.LOAN_DAYS and item.kind() without caring whether it holds a Book or a Magazine.
Property	**is_available** is a read-only property computed from available_copies.
Dictionaries	items, members and each member's borrowed dictionary (item_id to due date) give fast lookup.
date / timedelta	Used to compute due dates and the number of late days.

3. Class design

Class	Attributes	Methods
LibraryItem (parent)	item_id, title, total_copies, available_copies, LOAN_DAYS=14, FINE_PER_DAY=2	is_available (property), status(), details(), kind()
Book (child)	author (extra)	kind() overridden
Magazine (child)	issue_no (extra), LOAN_DAYS=7, FINE_PER_DAY=1	kind() overridden
Member	member_id, name, borrowed {id: due date}, fines_paid, MAX_ITEMS=2	-
Library	name, items, members, log	add_item(), add_member(), issue_item(), return_item(), search(), show_catalogue()

4. Algorithm

Issue an item

1. Check that the member and the item exist (otherwise raise LibraryError).
2. Reject if the member already holds the same item.
3. Reject if the member already holds MAX_ITEMS (2) items.

4. Reject if available_copies is zero.
5. Reduce available_copies by one, store due date = issue date + LOAN_DAYS, and log the issue.

Return an item

1. Reject if the member never borrowed that item.
2. Remove it from the member's borrowed dictionary and increase available_copies by one.
3. late days = max(0, return date - due date); fine = late days x FINE_PER_DAY.
4. Log the return and report the fine, if any.

5. Program

ex15_library.py

```
"""
Lab Exercise 15 : Library Management System
Concepts        : Classes, objects, inheritance, polymorphism,
                  availability tracking, issue / return logic, fines
"""

from datetime import date, timedelta


class LibraryError(Exception):
    """Base class for library related errors."""


# ---------------------------------------------------------------
# Base class : LibraryItem  (common attributes of everything we lend)
# ---------------------------------------------------------------
class LibraryItem:
    LOAN_DAYS = 14            # default loan period
    FINE_PER_DAY = 2          # Rs. per late day

    def __init__(self, item_id, title, copies=1):
        self.item_id = item_id
        self.title = title
        self.total_copies = copies
        self.available_copies = copies

    @property
    def is_available(self):
        return self.available_copies > 0

    def status(self):
        return "Available" if self.is_available else "Issued"

    def details(self):
        copies = f"{self.available_copies}/{self.total_copies}"
        return f"{self.item_id:<6}{self.title:<34}{self.kind():<10}{copies:<5} {self.status()}"

    def kind(self):
        return "Item"


# ---------------------------------------------------------------
# Derived classes : Book and Magazine (INHERITANCE)
# ---------------------------------------------------------------
class Book(LibraryItem):
    def __init__(self, item_id, title, author, copies=1):
        super().__init__(item_id, title, copies)
        self.author = author

    def kind(self):
        return "Book"


class Magazine(LibraryItem):
    LOAN_DAYS = 7             # method/attribute overriding: shorter loan period
    FINE_PER_DAY = 1

    def __init__(self, item_id, title, issue_no, copies=1):
        super().__init__(item_id, title, copies)
        self.issue_no = issue_no

    def kind(self):
        return "Magazine"
```

```python


# ---------------------------------------------------------------
# Member class
# ---------------------------------------------------------------
class Member:
    MAX_ITEMS = 2

    def __init__(self, member_id, name):
        self.member_id = member_id
        self.name = name
        self.borrowed = {}          # item_id -> due_date
        self.fines_paid = 0

    def __str__(self):
        return f"{self.member_id} ({self.name})"


# ---------------------------------------------------------------
# Library class : controls the whole system
# ---------------------------------------------------------------
class Library:
    def __init__(self, name):
        self.name = name
        self.items = {}
        self.members = {}
        self.log = []

    # ----- set-up -----
    def add_item(self, item):
        self.items[item.item_id] = item

    def add_member(self, member):
        self.members[member.member_id] = member

    # ----- lookup helpers -----
    def _item(self, item_id):
        if item_id not in self.items:
            raise LibraryError(f"Item '{item_id}' does not exist in catalogue.")
        return self.items[item_id]

    def _member(self, member_id):
        if member_id not in self.members:
            raise LibraryError(f"Member '{member_id}' is not registered.")
        return self.members[member_id]

    # ----- core operations -----
    def issue_item(self, member_id, item_id, on_date):
        member, item = self._member(member_id), self._item(item_id)
        if item_id in member.borrowed:
            raise LibraryError(f"{member.name} already holds '{item.title}'.")
        if len(member.borrowed) >= Member.MAX_ITEMS:
            raise LibraryError(f"{member.name} reached the limit of {Member.MAX_ITEMS} items.")
        if not item.is_available:
            raise LibraryError(f"'{item.title}' is currently not available (all copies issued).")
        item.available_copies -= 1
        due = on_date + timedelta(days=item.LOAN_DAYS)
        member.borrowed[item_id] = due
        self.log.append((on_date, "ISSUE", member_id, item_id))
        return f"Issued '{item.title}' to {member.name}. Due on {due:%d-%b-%Y}."

    def return_item(self, member_id, item_id, on_date):
        member, item = self._member(member_id), self._item(item_id)
        if item_id not in member.borrowed:
            raise LibraryError(f"{member.name} has not borrowed '{item.title}'.")
        due = member.borrowed.pop(item_id)
        item.available_copies += 1
        late_days = max(0, (on_date - due).days)
        fine = late_days * item.FINE_PER_DAY
        member.fines_paid += fine
        self.log.append((on_date, "RETURN", member_id, item_id))
        if fine:
            return f"Returned '{item.title}' {late_days} day(s) late. Fine = Rs. {fine}."
        return f"Returned '{item.title}' on time. No fine."

    def search(self, keyword):
        keyword = keyword.lower()
        return [i for i in self.items.values() if keyword in i.title.lower()]

    def show_catalogue(self):
        print(f"  {'ID':<6}{'TITLE':<34}{'TYPE':<10}{'AVL':<5} STATUS")
        print("  " + "-" * 66)
        for item in self.items.values():
```

```
            print("  " + item.details())


# ------------------------------------------------------------------
# Test harness
# ------------------------------------------------------------------
def attempt(title, action):
    try:
        print(f"  [OK]      {title:<38} -> {action()}")
    except LibraryError as err:
        print(f"  [BLOCKED] {title:<38} -> {err}")


def main():
    lib = Library("Central Library")
    lib.add_item(Book("B101", "Python Crash Course", "Eric Matthes", copies=2))
    lib.add_item(Book("B102", "Introduction to Algorithms", "T. H. Cormen", copies=1))
    lib.add_item(Book("B103", "Artificial Intelligence", "Stuart Russell", copies=1))
    lib.add_item(Magazine("M201", "Data Science Monthly", issue_no=42, copies=1))
    lib.add_member(Member("S01", "Sriram"))
    lib.add_member(Member("S02", "Rahul"))
    lib.add_member(Member("S03", "Ananya"))

    day0 = date(2026, 10, 1)      # fixed date so the output is reproducible

    print("=" * 78)
    print(" 1. INITIAL CATALOGUE")
    print("=" * 78)
    lib.show_catalogue()

    print()
    print("=" * 78)
    print(" 2. ISSUING BOOKS")
    print("=" * 78)
    attempt("S01 issues B101", lambda: lib.issue_item("S01", "B101", day0))
    attempt("S02 issues B101 (2nd copy)", lambda: lib.issue_item("S02", "B101", day0))
    attempt("S03 issues B101 (no copies left)", lambda: lib.issue_item("S03", "B101", day0))
    attempt("S01 issues B102", lambda: lib.issue_item("S01", "B102", day0))
    attempt("S01 issues B103 (limit of 2 reached)", lambda: lib.issue_item("S01", "B103", day0))
    attempt("S03 issues M201 (magazine, 7 days)", lambda: lib.issue_item("S03", "M201", day0))
    attempt("S01 issues B101 again (duplicate)", lambda: lib.issue_item("S01", "B101", day0))

    print()
    print("=" * 78)
    print(" 3. INVALID REQUESTS")
    print("=" * 78)
    attempt("Unknown member S99", lambda: lib.issue_item("S99", "B103", day0))
    attempt("Unknown book B999", lambda: lib.issue_item("S02", "B999", day0))

    print()
    print("=" * 78)
    print(" 4. CATALOGUE AFTER ISSUING")
    print("=" * 78)
    lib.show_catalogue()

    print()
    print("=" * 78)
    print(" 5. RETURNING (ON TIME AND LATE)")
    print("=" * 78)
    attempt("S01 returns B101 on day 10 (on time)", lambda: lib.return_item("S01", "B101", day0 + timedelta(days=10)))
    attempt("S02 returns B101 on day 20 (6 late)", lambda: lib.return_item("S02", "B101", day0 + timedelta(days=20)))
    attempt("S03 returns M201 on day 12 (5 late)", lambda: lib.return_item("S03", "M201", day0 + timedelta(days=12)))
    attempt("S03 returns B103 (never borrowed)", lambda: lib.return_item("S03", "B103", day0 + timedelta(days=12)))
    attempt("S03 issues B103 (after returns)", lambda: lib.issue_item("S03", "B103", day0 + timedelta(days=13)))

    print()
    print("=" * 78)
    print(" 6. SEARCH AND FINAL STATUS")
    print("=" * 78)
    print("  Search 'python' ->", [b.title for b in lib.search("python")])
    print("  Search 'data'   ->", [b.title for b in lib.search("data")])
    print()
    lib.show_catalogue()
    print()
    print("  Fines collected :", {m.name: m.fines_paid for m in lib.members.values()})
    print("  Transactions    :", len(lib.log))


if __name__ == "__main__":
    main()
```

6. Output

TERMINAL OUTPUT - python ex15_library.py

```
================================================================================
 1. INITIAL CATALOGUE
================================================================================
  ID    TITLE                          TYPE      AVL   STATUS
  ------------------------------------------------------------------
  B101  Python Crash Course            Book      2/2   Available
  B102  Introduction to Algorithms     Book      1/1   Available
  B103  Artificial Intelligence        Book      1/1   Available
  M201  Data Science Monthly           Magazine  1/1   Available

================================================================================
 2. ISSUING BOOKS
================================================================================
  [OK]      S01 issues B101                     -> Issued 'Python Crash Course' to Sriram. Due on 15-Oct-2026.
  [OK]      S02 issues B101 (2nd copy)          -> Issued 'Python Crash Course' to Rahul. Due on 15-Oct-2026.
  [BLOCKED] S03 issues B101 (no copies left)    -> 'Python Crash Course' is currently not available (all copies
            issued).
  [OK]      S01 issues B102                     -> Issued 'Introduction to Algorithms' to Sriram. Due on
            15-Oct-2026.
  [BLOCKED] S01 issues B103 (limit of 2 reached)-> Sriram reached the limit of 2 items.
  [OK]      S03 issues M201 (magazine, 7 days)  -> Issued 'Data Science Monthly' to Ananya. Due on 08-Oct-2026.
  [BLOCKED] S01 issues B101 again (duplicate)   -> Sriram already holds 'Python Crash Course'.

================================================================================
 3. INVALID REQUESTS
================================================================================
  [BLOCKED] Unknown member S99                  -> Member 'S99' is not registered.
  [BLOCKED] Unknown book B999                   -> Item 'B999' does not exist in catalogue.

================================================================================
 4. CATALOGUE AFTER ISSUING
================================================================================
  ID    TITLE                          TYPE      AVL   STATUS
  ------------------------------------------------------------------
  B101  Python Crash Course            Book      0/2   Issued
  B102  Introduction to Algorithms     Book      0/1   Issued
  B103  Artificial Intelligence        Book      1/1   Available
  M201  Data Science Monthly           Magazine  0/1   Issued

================================================================================
 5. RETURNING (ON TIME AND LATE)
================================================================================
  [OK]      S01 returns B101 on day 10 (on time)  -> Returned 'Python Crash Course' on time. No fine.
  [OK]      S02 returns B101 on day 20 (6 late)   -> Returned 'Python Crash Course' 6 day(s) late. Fine = Rs. 12.
  [OK]      S03 returns M201 on day 12 (5 late)   -> Returned 'Data Science Monthly' 5 day(s) late. Fine = Rs. 5.
  [BLOCKED] S03 returns B103 (never borrowed)     -> Ananya has not borrowed 'Artificial Intelligence'.
  [OK]      S03 issues B103 (after returns)       -> Issued 'Artificial Intelligence' to Ananya. Due on
            28-Oct-2026.

================================================================================
 6. SEARCH AND FINAL STATUS
================================================================================
  Search 'python' -> ['Python Crash Course']
  Search 'data'   -> ['Data Science Monthly']

  ID    TITLE                          TYPE      AVL   STATUS
  ------------------------------------------------------------------
  B101  Python Crash Course            Book      2/2   Available
  B102  Introduction to Algorithms     Book      0/1   Issued
  B103  Artificial Intelligence        Book      0/1   Issued
  M201  Data Science Monthly           Magazine  1/1   Available

  Fines collected : {'Sriram': 0, 'Rahul': 12, 'Ananya': 5}
  Transactions    : 8
```

7. Analysis of availability and issue / return conditions

Catalogue at the start: Python Crash Course (2 copies), Introduction to Algorithms (1), Artificial Intelligence (1), Data Science Monthly magazine (1). The simulation starts on 01-Oct-2026.

#	Operation	Condition tested	Result
1	S01 issues B101	Copy available	Issued, due 15-Oct-2026 (14 days)

#	Operation	Condition tested	Result
2	S02 issues B101	Second copy available	Issued; availability now 0/2
3	S03 issues B101	No copy left	BLOCKED: not available
4	S01 issues B102	Within limit of 2	Issued
5	S01 issues B103	Borrowing limit reached	BLOCKED: limit of 2 items
6	S03 issues magazine M201	Overridden loan period	Issued, due 08-Oct-2026 (7 days)
7	S01 issues B101 again	Duplicate issue	BLOCKED: already holds the book
8	Unknown member S99 / book B999	Invalid ids	BLOCKED: not registered / not in catalogue
9	S01 returns B101 on day 10	Before due date	Returned, no fine
10	S02 returns B101 on day 20	6 days late x Rs. 2	Returned, fine Rs. 12
11	S03 returns M201 on day 12	5 days late x Rs. 1	Returned, fine Rs. 5
12	S03 returns B103	Never borrowed	BLOCKED: has not borrowed
13	S03 issues B103 after returns	Copy available again	Issued, due 28-Oct-2026

Observation: the catalogue before and after the operations shows the availability changing correctly. B101 went from 2/2 to 0/2 and back to 2/2 after both returns. The final catalogue shows B102 and B103 issued (0/1), while B101 and M201 are available again. Total fines collected were Rs. 17 (Rahul Rs. 12, Ananya Rs. 5).

8. How the system supports book tracking and efficient resource management

- **Real-time availability.** available_copies is updated inside issue_item() and return_item(), so the catalogue always shows the true status without manual counting.
- **Tracking who has what.** Every member keeps a dictionary of borrowed items with due dates, and the Library keeps a log of every issue and return with the date, member and item.
- **Fair use of limited copies.** The limit of 2 items per member and the block on duplicate issue stop one person from holding many copies while others wait.
- **Faster turnover through fines.** Late fines encourage on-time returns, so popular books become available sooner.
- **Easy extension through inheritance.** A new type such as Journal or DVD only needs a small child class with its own loan period and fine rate; the issue and return logic does not change.
- **Quick lookup.** Dictionary storage gives direct access by id, and search() lets a user find titles by keyword.

9. Requirement checklist

Requirement in the question	Where it is solved
Classes and objects for books, issue and return	LibraryItem, Book, Magazine, Member, Library
Maintain availability status	available_copies, is_available, status()
Identify suitable classes and attributes	Section 3
Apply inheritance	Book and Magazine inherit from LibraryItem (with overriding)
Test availability and issue / return conditions	Output sections 2, 3 and 5; Section 7 table
Simulate library operations	Complete run shown in the output
Demonstrate book tracking and resource management	Catalogue views, borrowed dictionaries, log, Section 8

10. Conclusion

The library system correctly issues and returns books and magazines, keeps availability up to date, enforces borrowing rules, calculates fines, and handles invalid requests without crashing. Inheritance removed duplicate code and made the design easy to extend.

16 Dice Simulator using random module

Lab Exercise 16 | Simulation, frequency analysis, probability

QUESTION

Dice Simulator Using Random Module: Develop a Python program that simulates rolling one or more dice using Python's random module. Allow repeated rolls and display the outcomes.

- Select suitable random functions, design the simulation logic, test repeated trials, and analyze the frequency of possible outcomes.
- Relate the simulation to board games and probability experiments; interpret the observed outcomes across multiple rolls.

1. Aim

To simulate fair dice with Python's random module, roll one or several dice many times, display every outcome, count how often each result appears, and compare the observed frequencies with the theoretical probabilities.

2. Selecting suitable random functions

Function	What it does	Used here?
random.randint(a, b)	Random integer from a to b, **both included**. Matches a die: randint(1, 6).	**Yes** - main function for rolling
random.seed(n)	Fixes the starting state so the same sequence of rolls can be reproduced.	**Yes** - repeatable demonstration
random.choice(seq)	Picks one element of a sequence, e.g. choice([1,2,3,4,5,6]).	Alternative (same result)
random.random()	Float between 0 and 1; needs extra arithmetic to become a die face.	Not needed
random.shuffle / sample	Reorder or sample without repeats; useful for cards, not dice, because dice can repeat.	Not suitable

3. Simulation logic

1. Class **Die** stores the number of sides and has roll(), which returns random.randint(1, sides).
2. **roll_dice(n)** creates n dice, rolls each one and returns the list of faces.
3. **simulate(trials, n)** repeats the roll many times and counts the total of each roll with collections.Counter.
4. **expected_probability()** counts all equally likely outcomes to get the exact theoretical probability.
5. **show_frequency()** prints the count, observed %, expected % and a text histogram.
6. The main program runs: repeated single rolls, two-dice board-game turns, frequency tables for different trial sizes, a probability experiment and a small race game. Running with --play gives an interactive roller.

4. Program

ex16_dice.py

```
"""
Lab Exercise 16 : Dice Simulator using the random module
Concepts        : random.randint, random.seed, loops, dictionaries,
                  frequency analysis, probability, simulation
"""

import random
import sys
from collections import Counter


# ------------------------------------------------------------
```

```python
# Class Die : one fair die with a configurable number of sides
# ---------------------------------------------------------------
class Die:
    def __init__(self, sides=6):
        if sides < 2:
            raise ValueError("A die needs at least 2 sides.")
        self.sides = sides

    def roll(self):
        return random.randint(1, self.sides)      # both ends inclusive


# ---------------------------------------------------------------
# Simulator helpers
# ---------------------------------------------------------------
def roll_dice(num_dice=1, sides=6):
    """Roll num_dice dice once and return the list of faces."""
    dice = [Die(sides) for _ in range(num_dice)]
    return [d.roll() for d in dice]


def simulate(trials, num_dice=1, sides=6):
    """Roll the dice many times; return a Counter of the TOTAL of each roll."""
    return Counter(sum(roll_dice(num_dice, sides)) for _ in range(trials))


def expected_probability(total, num_dice=1, sides=6):
    """Exact theoretical probability of a total (by counting all outcomes)."""
    ways = {0: 1}
    for _ in range(num_dice):
        new = {}
        for s, w in ways.items():
            for face in range(1, sides + 1):
                new[s + face] = new.get(s + face, 0) + w
        ways = new
    return ways.get(total, 0) / (sides ** num_dice)


def show_frequency(counter, trials, num_dice=1, sides=6):
    """Print frequency table with a text histogram and theory comparison."""
    print(f"  {'Total':<7}{'Count':>7}{'Observed%':>11}{'Expected%':>11}   Histogram")
    print("  " + "-" * 70)
    for total in range(num_dice, num_dice * sides + 1):
        count = counter.get(total, 0)
        obs = 100 * count / trials
        exp = 100 * expected_probability(total, num_dice, sides)
        bar = "#" * round(obs * 2)
        print(f"  {total:<7}{count:>7}{obs:>10.2f}%{exp:>10.2f}%   {bar}")


# ---------------------------------------------------------------
# Optional interactive mode:  python ex16_dice.py --play
# ---------------------------------------------------------------
def play():
    print("DICE SIMULATOR  (type q to quit)")
    while True:
        cmd = input("How many dice to roll? ").strip().lower()
        if cmd == "q":
            break
        if not cmd.isdigit() or int(cmd) < 1:
            print("Please enter a positive whole number.")
            continue
        faces = roll_dice(int(cmd))
        print("You rolled:", faces, "| Total =", sum(faces))


# ---------------------------------------------------------------
# Demonstration run (seeded so the results can be reproduced)
# ---------------------------------------------------------------
def main():
    random.seed(2026)

    print("=" * 78)
    print(" 1. SINGLE DIE - TEN REPEATED ROLLS")
    print("=" * 78)
    rolls = [roll_dice()[0] for _ in range(10)]
    print("  Outcomes :", rolls)
    print("  Average  :", sum(rolls) / len(rolls), " (theoretical mean = 3.5)")

    print()
    print("=" * 78)
    print(" 2. TWO DICE - FIVE ROLLS (BOARD-GAME STYLE)")
    print("=" * 78)
```

```
    for turn in range(1, 6):
        d1, d2 = roll_dice(2)
        extra = "  <- DOUBLE! roll again" if d1 == d2 else ""
        print(f"  Turn {turn}: {d1} + {d2} = {d1 + d2}{extra}")

    print()
    print("=" * 78)
    print(" 3. FREQUENCY ANALYSIS - ONE DIE, DIFFERENT NUMBER OF TRIALS")
    print("=" * 78)
    for trials in (60, 600, 60000):
        print(f"\n  Trials = {trials}")
        show_frequency(simulate(trials), trials)

    print()
    print("=" * 78)
    print(" 4. FREQUENCY ANALYSIS - TWO DICE, 36000 TRIALS")
    print("=" * 78)
    trials = 36000
    show_frequency(simulate(trials, num_dice=2), trials, num_dice=2)

    print()
    print("=" * 78)
    print(" 5. PROBABILITY EXPERIMENT - GETTING A SIX / DOUBLE SIX")
    print("=" * 78)
    n = 100000
    sixes = sum(1 for _ in range(n) if roll_dice()[0] == 6)
    double6 = sum(1 for _ in range(n) if roll_dice(2) == [6, 6])
    print(f"  P(six)        observed = {sixes / n:.4f}   theory = {1 / 6:.4f}")
    print(f"  P(double six) observed = {double6 / n:.4f}   theory = {1 / 36:.4f}")

    print()
    print("=" * 78)
    print(" 6. BOARD GAME SIMULATION - RACE TO SQUARE 50")
    print("=" * 78)
    players = {"Sriram": 0, "Rahul": 0}
    turn = 0
    while max(players.values()) < 50:
        turn += 1
        for name in players:
            roll = roll_dice()[0]
            players[name] = min(50, players[name] + roll)
            if players[name] == 50:
                print(f"  Turn {turn:>2}: {name} rolled {roll} and reached square 50. {name} WINS!")
                break
        else:
            print(f"  Turn {turn:>2}: " + " | ".join(f"{k} at {v:>2}" for k, v in players.items()))
            continue
        break


if __name__ == "__main__":
    if "--play" in sys.argv:
        play()
    else:
        main()
```

5. Output

TERMINAL OUTPUT - python ex16_dice.py

```
==============================================================================
 1. SINGLE DIE - TEN REPEATED ROLLS
==============================================================================
  Outcomes : [1, 3, 5, 5, 6, 1, 2, 5, 5, 5]
  Average  : 3.8  (theoretical mean = 3.5)

==============================================================================
 2. TWO DICE - FIVE ROLLS (BOARD-GAME STYLE)
==============================================================================
  Turn 1: 4 + 5 = 9
  Turn 2: 5 + 6 = 11
  Turn 3: 4 + 5 = 9
  Turn 4: 4 + 2 = 6
  Turn 5: 1 + 5 = 6

==============================================================================
 3. FREQUENCY ANALYSIS - ONE DIE, DIFFERENT NUMBER OF TRIALS
==============================================================================
```

```
  Trials = 60
  Total    Count  Observed%  Expected%   Histogram
  ------------------------------------------------------------------------
  1           10     16.67%     16.67%   ##############################
  2            2      3.33%     16.67%   #######
  3           12     20.00%     16.67%   ####################################
  4           14     23.33%     16.67%   ##########################################
  5           13     21.67%     16.67%   #######################################
  6            9     15.00%     16.67%   ###########################

  Trials = 600
  Total    Count  Observed%  Expected%   Histogram
  ------------------------------------------------------------------------
  1           93     15.50%     16.67%   ###############################
  2           93     15.50%     16.67%   ###############################
  3          107     17.83%     16.67%   ####################################
  4           98     16.33%     16.67%   #################################
  5          105     17.50%     16.67%   ###################################
  6          104     17.33%     16.67%   ###################################

  Trials = 60000
  Total    Count  Observed%  Expected%   Histogram
  ------------------------------------------------------------------------
  1        10177     16.96%     16.67%   ##################################
  2         9992     16.65%     16.67%   #################################
  3        10134     16.89%     16.67%   ##################################
  4        10017     16.70%     16.67%   #################################
  5         9863     16.44%     16.67%   #################################
  6         9817     16.36%     16.67%   #################################

================================================================================
 4. FREQUENCY ANALYSIS - TWO DICE, 36000 TRIALS
================================================================================
  Total    Count  Observed%  Expected%   Histogram
  ------------------------------------------------------------------------
  2         1024      2.84%      2.78%   ######
  3         2037      5.66%      5.56%   ###########
  4         2953      8.20%      8.33%   ################
  5         4073     11.31%     11.11%   ######################
  6         4952     13.76%     13.89%   ###########################
  7         5932     16.48%     16.67%   ################################
  8         4991     13.86%     13.89%   ###########################
  9         3985     11.07%     11.11%   ######################
  10        3023      8.40%      8.33%   ################
  11        2024      5.62%      5.56%   ###########
  12        1006      2.79%      2.78%   ######

================================================================================
 5. PROBABILITY EXPERIMENT - GETTING A SIX / DOUBLE SIX
================================================================================
  P(six)        observed = 0.1664   theory = 0.1667
  P(double six) observed = 0.0284   theory = 0.0278

================================================================================
 6. BOARD GAME SIMULATION - RACE TO SQUARE 50
================================================================================
  Turn  1: Sriram at  5 | Rahul at  5
  Turn  2: Sriram at  7 | Rahul at  9
  Turn  3: Sriram at  9 | Rahul at 13
  Turn  4: Sriram at 12 | Rahul at 17
  Turn  5: Sriram at 18 | Rahul at 21
  Turn  6: Sriram at 22 | Rahul at 27
  Turn  7: Sriram at 25 | Rahul at 29
  Turn  8: Sriram at 29 | Rahul at 35
  Turn  9: Sriram at 35 | Rahul at 36
  Turn 10: Sriram at 40 | Rahul at 38
  Turn 11: Sriram at 43 | Rahul at 44
  Turn 12: Sriram at 45 | Rahul at 47
  Turn 13: Sriram rolled 6 and reached square 50. Sriram WINS!
```

6. Analysis of the frequency of outcomes

6.1 One die: effect of the number of trials

Trials	Expected for each face	Observed range	Largest gap from 16.67%
60	16.67%	3.33% (face 2) to 23.33% (face 4)	about 13.3 points
600	16.67%	15.50% to 17.83%	about 1.2 points
60,000	16.67%	16.36% to 16.96%	about 0.3 points

With only 60 rolls, face 2 appeared just twice, which looks unfair but is normal random variation. As the number of trials grew, every face moved closer to one sixth. This is the **law of large numbers**: the more times the experiment is repeated, the closer the observed frequency gets to the theoretical probability.

6.2 Two dice: totals are not equally likely

Total	2	3	4	5	6	7	8	9	10	11	12
Ways (of 36)	1	2	3	4	5	6	5	4	3	2	1
Theory %	2.78	5.56	8.33	11.11	13.89	16.67	13.89	11.11	8.33	5.56	2.78
Observed % (36,000 rolls)	2.84	5.66	8.20	11.31	13.76	16.48	13.86	11.07	8.40	5.62	2.79

The histogram is a triangle with its peak at 7, because six of the 36 combinations give 7 (1+6, 2+5, 3+4, 4+3, 5+2, 6+1), while 2 and 12 can be made in only one way. The simulated values match the theory to within about 0.2 percentage points.

6.3 Probability experiment

In 100,000 trials the observed probability of a six was 0.1664 (theory 0.1667) and of a double six was 0.0284 (theory 0.0278), again very close to the exact values 1/6 and 1/36.

7. Relation to board games and probability experiments

- **Moving tokens.** In games like Snakes and Ladders, Ludo or Monopoly the dice decide how many squares a player moves. Section 6 of the output simulates a race to square 50 using exactly this idea.
- **Doubles.** Many games give a bonus or a penalty for rolling doubles (Monopoly's extra turn, jail after repeated doubles). The two-dice demonstration flags a double, and the experiment shows it occurs about 1 time in 6.
- **Needing a six.** Ludo traditionally needs a six to bring a token out. Since P(six) = 1/6, a player waits about six turns on average, which the simulation confirms.
- **Why 7 is special.** Games that use two dice (for example Catan) give number 7 the highest chance and numbers 2 and 12 the lowest, exactly as the frequency table shows.
- **Fairness testing.** Comparing observed and expected frequencies is how one can check whether a real die is biased.

8. Interpretation across multiple rolls

Short sequences are unpredictable: the first ten rolls averaged 3.8 although the true mean is 3.5, and a single face can appear too often or too rarely in a small sample. Each roll is independent, so past results do not influence the next roll. Only over many trials do the frequencies settle near the probabilities. Because the module generates pseudo-random numbers, fixing the seed reproduces the same results, which is useful for testing and for sharing results.

9. Requirement checklist

Requirement in the question	Where it is solved
Simulate rolling one or more dice with the random module	Die, roll_dice(n), random.randint
Allow repeated rolls and display outcomes	Output sections 1, 2 and 6; --play interactive mode
Select suitable random functions	Section 2 comparison table
Design the simulation logic	Section 3 and the program
Test repeated trials	60, 600, 60,000 and 36,000 trials, 100,000-trial experiment
Analyse the frequency of outcomes	Section 6 tables and histograms
Relate to board games and probability experiments	Section 7 and the race to square 50
Interpret observed outcomes across multiple rolls	Section 8

10. Conclusion

The simulator rolls any number of dice using random.randint, repeats rolls as often as needed and displays the results. The frequency analysis confirms that a single die is uniform, that two-dice totals follow a triangular distribution peaking at 7, and that observed results approach theoretical probabilities as the number of trials increases.

End of Lab Exercises 14, 15 and 16

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